

BUSINESS ANALYTICS TOOLS – II

Machine Learning Project Report

Customer Churn Prediction for a Subscription Company

Industry: Telecom

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1. Project Problem Definition & Business Understanding:

Customer churn prediction is one of the most important business analytics problems in subscription-based industries such as telecom, SaaS, and retail. Customer churn occurs when existing customers discontinue their services or subscriptions. High churn rates directly affect company revenue, increase customer acquisition costs, and reduce long-term profitability.

The primary objective of this project is to build a machine learning model capable of predicting customers who are likely to leave the company. By identifying high-risk customers in advance, organizations can take proactive actions such as personalized offers, loyalty programs, and customer support improvements.

Key **stakeholders** involved in this project include:

- Telecom companies and subscription-based businesses.
- Marketing and customer retention teams.
- Business managers and decision-makers.
- Customer relationship management (CRM) teams.
- Data analysts and machine learning engineers.

The **expected outcome of this project** is an intelligent churn prediction system that improves customer retention strategies and helps organizations reduce revenue loss.

2. Data Collection & Description:

The project uses the **Telco Customer Churn Dataset**, which contains customer demographics, account details, subscription information, and churn labels. The dataset is widely used for machine learning classification tasks related to customer retention.

Dataset Characteristics:

- Dataset Source: Telco Customer Churn Dataset
- Industry Domain: Telecom
- Target Variable: Churn
- Data Type: Structured tabular data
- Features include tenure, contract type, payment methods, internet service, and monthly charges.

Several data quality challenges were identified during preprocessing. These included missing values, categorical feature encoding requirements, and class imbalance between churned and non-churned customers.

3. Data Preprocessing & Feature Engineering

Data preprocessing is an essential step in machine learning because raw datasets often contain inconsistencies and irrelevant information. The following preprocessing techniques were applied in this project:

- Overseeing missing and null values.
- Encoding categorical variables into numerical format.
- Feature scaling and normalization.
- Train-test splitting for validation.
- Class imbalance handling using `scale_pos_weight` in XGBoost.

Feature engineering improved the predictive performance of the models by transforming raw data into meaningful numerical representations suitable for machine learning algorithms.

4. Model Development & Technology Stack

The project was implemented using Python and several machine learning libraries. Different classification algorithms were tested and compared to identify the most effective churn prediction model.

Technology Stack:

- Programming Language: Python
- Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost
- Development Environment: Jupyter Notebook

Machine Learning Models Used:

- Naive Bayes
- K-Nearest Neighbors (KNN)
- XGBoost

5. Model Evaluation & Results:

Model	Accuracy	F1 Score	ROC-AUC
Naive Bayes	67.01%	58.04%	81.01%
KNN	78.51%	55.84%	82.13%
XGBoost	75.06%	60.88%	82.38%

Interpretations:

i) Naive Bayes:

Accuracy: 67.01%

F1 Score: 58.04%

ROC-AUC Score: 81.01%

- Naive Bayes achieved moderate prediction performance.
- The model showed strong recall for churned customers, meaning it successfully identified many customers likely to leave.
- Overall accuracy was lower compared to other models.
- The model may oversimplify relationships because it assumes feature independence.

ii) K-Nearest Neighbors (KNN)

Accuracy: 78.51%

F1 Score: 55.84%

ROC-AUC Score: 82.13%

- KNN performed significantly better in terms of accuracy.
- The model predicted retained customers very effectively.
- KNN achieved the highest overall accuracy among the tested models.
- F1 score for churn prediction was lower than XGBoost, indicating weaker balance between precision and recall for churn detection.

iii) XGBoost

Accuracy: 75.06%

F1 Score: 60.88%

ROC-AUC Score: 82.38%

- XGBoost produced the best balance between accuracy and churn detection capability.
- It achieved the highest F1 Score and highest ROC-AUC score among all models.
- The model managed class imbalance effectively using `scale_pos_weight`.
- XGBoost was better at identifying actual churned customers compared to KNN and Naive Bayes.

Conclusion:

Although KNN achieved the highest raw accuracy, XGBoost provided the best balance between precision and recall. This makes XGBoost the most reliable model for real-world churn prediction because churn datasets are typically imbalanced.

The evaluation metrics demonstrated that machine learning techniques can effectively identify customers at risk of leaving. These insights can help organizations design targeted retention campaigns.

6. Business Interpretation of Results:

The findings of this project have strong business implications for telecom and subscription-based companies. By identifying high-risk customers early, businesses can reduce churn and improve customer satisfaction.

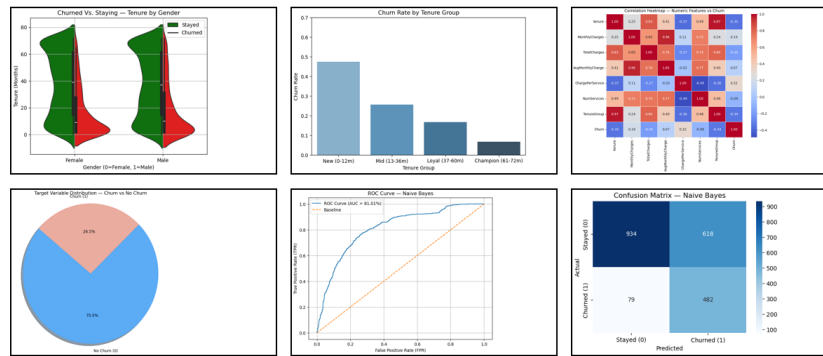
- Retention campaigns can target high-risk customers.
- Customer service teams can proactively engage users.
- Marketing teams can personalize promotional offers.
- Businesses can improve long-term customer loyalty.
- Predictive analytics supports data-driven decision-making.

7. Prototype / Dashboard / Deployment Blueprint:

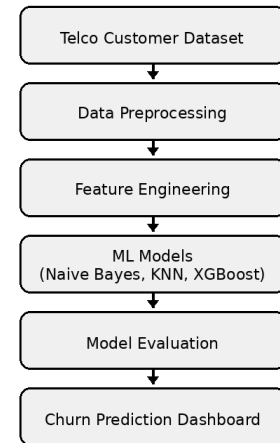
The project dashboard presents visualization results along with the system architecture diagram. The dashboard summarizes churn trends, customer behavior patterns, confusion matrices, ROC curves, and deployment workflow.

Customer Churn Prediction Dashboard & System Architecture

Machine Learning Project Visualization Results



System Architecture Diagram



Key Insights:

- XGBoost achieved the best churn prediction performance.
- KNN produced the highest accuracy but lower churn recall.
- Dashboard visualizations support business decision-making.
- Predictive analytics helps telecom companies reduce customer loss.

The deployment blueprint demonstrates how customer data flows through preprocessing, feature engineering, machine learning models, and final churn prediction dashboards.

8. Deployment Challenges:

- Managing real-time customer data updates.
- Managing class imbalance in future datasets.
- Ensuring model scalability for large telecom databases.
- Maintaining model accuracy over time.
- Data privacy and security concerns during deployment.

9. Retention Strategy Recommendations:

Based on the predictive analysis, several customer retention strategies are recommended for subscription companies.

- Provide personalized discounts to high-risk customers.
- Introduce loyalty reward programs.

- Improve customer service response times.
- Use predictive alerts for churn-risk customers.
- Conduct customer feedback analysis regularly.

10. Conclusion:

This project successfully demonstrated the application of machine learning techniques for customer churn prediction. The developed models provided valuable insights into customer behavior and churn patterns. Among all tested algorithms, XGBoost achieved the best overall performance and proved to be the most effective model for churn prediction. The developed churn prediction model successfully identifies customers who are likely to leave the company based on their subscription behavior, billing patterns, and service usage characteristics. The project also highlighted how predictive analytics can support business decision-making and improve customer retention strategies.